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263-3300-10L Data Science Lab: Predicting Nitrous Oxide (N_2O) Emissions From Agriculture

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Abstract

Nitrous oxide (N_2O) is the third most important greenhouse gas, with the majority of anthropogenic emissions originating from agricultural activities. However, large temporal and spatial variability in N_2O fluxes combined with limited field observations, continues to hinder accurate prediction and the identification of key emission drivers. In this study, we apply several machine learning models such as Random Forest, XGBoost, ElasticNet, Support Vector Machine and Multilayer Perceptron to predict N_2O emissions across multiple agricultural sites in Switzerland. To explicitly account for temporal dynamics, we construct lagged and rolling-lag features for key environmental and management variables. Model interpretability is achieved using SHAP values, which allow us to quantify the contribution of individual predictors to N_2O emissions. By comparing SHAP-based feature importance across sites, we identify both site-specific and consistent drivers of N_2O emissions. Our results highlight the dominant role of nitrogen availability and soil conditions, while also revealing substantial variability in driver importance across fields. These findings demonstrate the potential of interpretable machine learning to improve understanding of N_2O emission processes and inform targeted mitigation strategies.

Keywords

Data Science, XGBoost, Random Forest, Support Vector Machine, SHAP Values, Neural Networks, Environmental Science, Climate Change, Greenhouse Gases

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Author contributions using the CRediT framework [? ?]:
AD: Methodology, Software, Visualization, Investigation, Writing - Original Draft
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FT, LA: Conceptualization, Supervision
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1 Introduction

Nitrous oxide (N_2O) plays a critical role in climate change due to its high global warming potential. As N_2O is more than 300 times more effective in trapping heat compared to carbon dioxide [1], improving our understanding of the drivers of N_2O is critical for effective mitigation. Agriculture represents a dominant source of N_2O emissions, primarily through nitrogen fertilization and soil management practices [1]. However, predicting N_2O emissions from agricultural fields remains challenging due to high temporal variability, occurrences of episodic high emission events, and the inherently noisy nature of environmental and flux measurements. In addition, strong field specific behaviour further limits the robustness and generalizability of predictive models [2].

As a result, several studies have turned to the use of various types of machine learning models to predict N_2O in agricultural fields [3–7]. While these models demonstrate moderate predictive capabilities, they are often treated as black boxes, limiting their utility for interpretation. To address this, some studies have incorporated SHAP values in order to improve model interpretability [8]. In this study, we combine explicit temporal feature engineering with multiple machine learning models to predict N_2O emissions across several agricultural sites around Switzerland. To identify influential predictors, we employ SHAP-based analysis to assess the robustness and consistency of the importance of variables across datasets and model configurations.

2 Related Work

The methods used for predicting N_2O emissions have rapidly evolved over the past few years. Although initial attempts focused on leveraging simple linear regression [9] and biological process-based modelling [10], these approaches often lacked the ability to capture the non-linear nature of N_2O flux data whilst also requiring site-specific calibration.

These shortcomings quickly gave rise to the adoption of machine-learning approaches for the predicting N_2O emissions. Earlier studies, such as that of Saha et al. (2021), leveraged models such as Random Forest due to their ability to capture complex non-linear relationships, achieving a R^2 of 0.38 [7]. On the other hand, more recent studies, such as that of Gnisia et al. (2025), experimented with both "shallow" models such as XGBoost and Random Forest, as well as deep learning models such as feed-forward neural networks, and found that the "shallower" models generally perform better for frequent flux prediction such as weekly fluxes, whereas deep learning models excel at predicting less frequent fluxes such as annual fluxes [6].

In our study, we look to build on these previous studies by exploring both commonly used models such as XGBoost Random Forest as well as alternative approaches that haven't been experimented with in as much detail, such as Support Vector Machine and Multilayer Perceptron.

3 Background

3.1 Datasets

The data used in this study is a mixture of open-source data and data that is yet to be published. The data was collected by the Grassland sciences group at ETH in the last decade at different agricultural sites in Switzerland. We explored 6 different datasets which were either cropland, grassland or cropland with a temporary period of grassland. These datasets varied in length, with Chamau 2005-2024 [11] being over the longest period of time, whereas the other datasets (Oensingen 2018-19 [12], Aeschi 2018-20 [12, 13], Oensingen 2021-23 [14], Forel 2024-25 (unpublished) and Tänikon 2023-25 (unpublished)) covered much shorter time spans of around 1-2 years.

3.2 Variables

Each dataset consists of half-hourly N_2O emissions, as well as meteorological, environmental, and agronomic management data from multiple years. For the Chamau and Tänikon datasets, a parcel variable was also provided, since the land was divided into two adjacent field parcels that followed different management strategies.

The half-hourly N_2O emissions represented our target variable, and, due to fairly large numbers of missing values for this variable, the datasets also contained additional columns with imputed values for N_2O emissions using Random Forest and XGBoost.

The meteorological and environmental variables were all measured on a continuous scale and included predictors such as net flux of CO_2 from the field to the atmosphere (with positive values indicating CO_2 was released into the atmosphere and negative values indicating CO_2 was absorbed by the field), precipitation (measured in millimeters), soil water content (measured as a percentage), air temperature and soil temperature (measured in degrees Celsius).

The agronomic management data consisted of management events such as mowing, sowing, soil cultivation, and fertilizer applications, which were provided as binary indicators. In the case of fertilizer applications, we were also provided with additional datasets containing information on the dates and magnitude of nitrogen fertilization, and for the Chamau and Tänikon sites, these datasets also included parcel specific information.

4 Methodology

4.1 Data Preprocessing

For each dataset, the raw data was first pruned by selecting only the relevant meteorological, management, and flux variables, followed by renaming columns to enforce common names across all datasets.

In order to capture the temporal influence of management events, we calculated a "days-since" variable for each management event which counts the number of days since this event last occurred, with values being capped at 30 days as events older than this have negligible remaining effect on N_2O emissions. In the case of the Chamau and Tänikon datasets, management events were done separately on two adjacent field parcels. As the eddy-covariance sensor was positioned between the two parcels, each observation in the base dataset needed to be assigned using the wind direction relative to the sensor. For Chamau, observations with wind direction between 110°–210° to allocated to field A, and rows between 290°–360° and 0°–30° to field B. These wind directions were chosen based on the findings of Feigenwinter et al. (2023) while working on the same dataset [15]. For the fields that did not fall into either category, we labeled them as “Uncertain” and defined parcel-specific management variables for these rows as the average of the corresponding A and B values. In the case of the Tänikon datasets, parcel information was directly provided in the raw data through the ‘parcel‘ and ‘parcel_certainty‘ columns.

To capture the temporal nature of the continuous variables, we generated a large set of lagged meteorological and environmental features. The base data (which often came at frequent intervals throughout the day) was first aggregated to daily totals for precipitation and daily means for all other variables. For each observation in the dataset, we added the observed daily averages (or sum) of 1, 3, 5, and 7 calendar days prior to the timestamp of the observation. To better capture temporal patterns, we additionally computed rolling-lags which took both the sum and the mean of the daily averages (or sum) for the previous 3, 5 or 7 calendar days. After these lagged variables were generated for each dataset, they were joined back to the base dataset by calendar date. For the datasets with parcels, these lags were calculated separately per parcel, and joined back to the base dataset by calendar date and parcel.

The ancillary datasets containing detailed information on the magnitude of nitrogen fertilization events were joined with the original data using the date and, if applicable, parcel. Once integrated, we computed nitrogen decay variables using the 3, 5, and 7 day half lives and added them to the base dataset as additional temporal variables. For each half-life h and each day t, we compute an exponentially decayed nitrogen stock using

$$S_t^{(h)} \;=\; D_t \;+\; \left(0.5^{\frac{\Delta t}{h}}\right) S_{t-1}^{(h)},$$

where

$$\Delta t = \text{number of days between timestamps } t \text{ and } t-1,$$

D_t is the fertilizer amount applied on day t, and $S_t^{(h)}$ is the accumulated decayed nitrogen remaining on day t given a half-life of h days.

Missing days in the data are handled naturally because Δt may exceed 1, leading to proportionally greater decay via the factor $0.5^{\Delta t/h}$.

Finally, the target N_2O flux was log-transformed in the form

$$N_2O\_Flux\_ln = \log(1 + N_2O\_Flux)\,.$$

with non-positive values set to zero. This log transformation was done since N_2O fluxes are often highly skewed with a heavy right tail. In doing this transformation, we found that it significantly increased results by reducing the impact and variance of the data from extremely high outlier events.

All steps previously outlined were used to construct our hourly-level dataset. However, to create a daily dataset, we aggregated all variables by calendar date and parcel (if applicable). Continuous meteorological and environmental flux variables were aggregated using the daily mean, while precipitation was aggregated using the total daily sum. Management events were encoded using the daily maximum to indicate whether an event occurred on that calendar day, and their corresponding "days-since-event" variables were aggregated using the daily minimum to preserve the closest proximity to the last event. As the lag variables capture changes in the hourly dataset by the lag of the previous calendar dates, these lagged variables can be seamlessly integrated into the daily aggregated dataset, as these variables are the same for all observations with the same calendar date.

4.2 Models

Given the two types of data we constructed (hourly and daily), we decided to design a series of experiments using several different machine learning models. For both the hourly and daily data we used 3 different experimental settings, namely **base**, **lag** and **augmented lag**, where we increasingly included more predictors that the models could use in order to better predict N_2O emissions. The **base** setting includes only the predictors included in the original data, the **lag** setting includes all lag variables except the rolling means/sums and finally the **augmented lag** setting includes all variables derived in section 4.1. In terms of models, we have selected the following five for our experiments:

(1) **ElasticNet:** after some discussions with our academic coach, we have opted to include this model as the baseline for our experiments.
(2) **Random Forest:** this model has been widely used in literature due to its ability to handle complex, non-linear data with a large number of variables [3, 7], as is the case with our datasets.
(3) **Extreme Gradient Boosting (XGBoost):** this model has also been widely used for predicting N_2O fluxes not only in agriculture [6] but also in municipal wastewater treatment plants [4] .
(4) **Support Vector Machine (SVM):** we have opted for this model also due to its ability to capture complex, non-linear relationships in a dataset, as well as due to very recent studies in literature also showing that this model achieves promising results [5].
(5) **Multilayer Perceptron (MLP):** recent studies such as that of Gnisia et al. (2025) have shown that deep learning models, although weaker than less complex models on tasks such as weekly N_2O flux prediction, achieve promising results for predicting annual N_2O fluxes [6]. We have therefore opted to experiment with this model to observe its performance in hourly and daily settings.

All of these models were implemented using the scikit-learn Python library [16].

4.3 Hyperparameter optimisation

After obtaining some initial results on the five models previously outlined, we selected the three most promising ones and performed hyperparameter optimisation to see if their accuracy could be further improved. The optimisation setup we used was Bayesian Optimisation [17] with 500 iterations, which we implemented using the hyperopt Python library [18]. Since some of the models may be prone to overfitting and could therefore show worse performance after running our optimisation pipeline, we implemented an additional check before saving the best parameters, where we compare the reported R^2 for the base and optimised model and, for each experiment, we store the parameters of the model that achieved a higher R^2 during testing.

We include a detailed breakdown of the hyperparameter search space used for each model in Appendix C.

4.4 Data Splitting & Validation Strategy

The split we used between our training and test data was a time-based split, meaning in each experiment our test set was from a much later time period than our training data. In the case of **Random Forest**, the split we used was **80:20**, whereas for **XGBoost**, **MLP** and **SVM** the split we used was **90:10**. We have opted for these splits after experimenting with 5 different split settings for each model, namely 70:30, 85:15, 80:20, 90:10, 75:25, and choosing the split setting that offered us the highest R^2.

For the experiments involving hyperparameter optimisation, the splits used between train, validation and test sets were **65:15:20** for **Random Forest**, and **80:10:10** for **XGBoost** and **SVM**. In these experiments we also used k-fold cross validation [19] using 9 folds for **XGBoost** and only 3 folds for **Random Forest** and **SVM** for computational feasibility.

In addition to our data splitting setup, we also incorporated a data leakage check in each of our experiments to ensure the validity of our results. This check looks at the dates in the train and test datasets and displays the number of dates (if any) that appear in both datasets, thus ensuring that there is no overlap between the two.

4.5 Explainability

To provide explainability for our Random Forest and XGBoost models, we employ **Shapeley Additive Explanation (SHAP) values** [20], which identify predictors that are most strongly associated with model predictions by quantifying how much each variable increases or decreases a model's prediction relative to an average baseline, while also accounting for interactions with other features.

We computed SHAP values for both XGBoost and Random Forest for our 2 best performing datasets (Chamau [11] and Oensingen 2021-23 [14]). The SHAP values were calculated on the held-out test set based on our train-test split described in section 4.3. Global feature importance is quantified as the mean absolute SHAP values across all test observations, and beeswarm plots are used to visualize the magnitude and direction of the feature effects on predicted N_2O emissions.

To identify the common drivers of N_2O emissions across the different datasets and configurations, we aggregate SHAP-based feature rankings from both XGBoost and Random Forest. For each model, we rank predictors by their mean absolute SHAP values and the top-K features ($K = 3, 5, 10$) are extracted. Models are first grouped by experimental setting (base, lag, augmented lag) and within each group we compute how frequently each feature appears among the top-K ranked predictors. To summarize these results, we report the percentage of settings in which each predictor appears in the top-K ranked features. Thus, a feature with a higher percentage is more likely to be a robust predictor of N_2O emissions across different fields.

5 Results

5.1 Experiments

Among all datasets, the highest predictive performance in terms of R^2 was achieved on the Chamau and Oensingen 2021–23 datasets. The best result across all models and feature settings was the Random Forest model trained on the Oensingen 2021–23 dataset under the daily base setting, achieving a R^2 of 0.531. For the Chamau dataset, the highest performance was obtained using the Random Forest model under the daily lag setting, with an R^2 of 0.371. The full set of R^2 values for all models, datasets, and experimental configurations is reported in the Appendix B.

Figure 1 shows observed and predicted N_2O flux time series for our best performing model, namely Random Forest. Across both datasets, the model captures broad temporal trends and seasonal changes in N_2O emissions, while smoothing short-term variability and extreme peaks. In the best performing Chamau setting, peaks in observed N_2O emissions occasionally coincide with fertilization events, and the model correspondingly predicts elevated emissions following these interventions. This suggests that the model has learned a temporal association between fertilization and increased N_2O fluxes. However, this learned association can sometimes be misleading, as not all fertilization events result in elevated emissions. In such cases, the model still predicts increased fluxes despite the absence of a corresponding observed response, highlighting a limitation in its ability to distinguish between fertilization events that trigger emission pulses and those that do not.

In the best performing Oensingen 2021-23 setting, the model generally captures the timing of emission spikes prior to the first fertilization event, but subsequently underestimates the magnitude of the observed increase. One possible explanation is that, although fertilization events are recorded in the dataset, no information on the applied nitrogen amount was available for all such events, thereby limiting the model's ability to distinguish between high and low fertilization events. As no nitrogen fertilizer magnitude datapoints are present in the test set, this limitation likely contributes to the stronger performance observed for models trained on the base feature set, as the half-life decay of nitrogen fertilizer cannot be used to predict emissions.

Across experiments, models trained on daily aggregated data consistently outperformed those trained on higher-frequency data. This improvement is likely attributable to the aggregation process reducing diurnal noise and extreme episodic fluctuations in the N_2O flux measurements. In terms of models, tree-based ensemble methods such as Random Forest and XGBoost consistently outperformed linear and neural network models in predicting N_2O emissions. ElasticNet, while providing a useful baseline, showed limited predictive skill, indicating that linear relationships alone are insufficient to capture the complex and nonlinear dynamics governing N_2O fluxes. Support Vector Machines and Multilayer Perceptrons exhibited mixed performance across datasets and temporal resolutions, and while these models occasionally achieved competitive results, their performance was less stable and more sensitive to hyperparameter choices and data availability, particularly in smaller datasets. Overall, Random Forest and XGBoost emerged as the most robust modeling approaches, offering a favourable balance between predictive performance and stability across sites and feature configurations.

5.2 SHAP Analysis

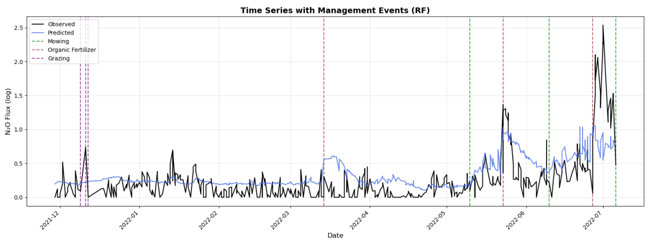
Given our experimental results, we have only opted to use the Random Forest and XGBoost models for our SHAP value analysis as they were our best performing and most consistent models. As such, we present SHAP beeswarm plots in Figure 2 for these two models trained on the Chamau and Oensingen 2021-23 datasets. For each dataset/model combination, the configuration that achieved the highest predictive performance (in terms of R^2) was shown.

In each beeswarm plot, features are ordered vertically by their mean absolute SHAP value, such that variables appearing higher in the plot exert a larger overall influence on the model predictions. Each point corresponds to a single observation in the test set, with its horizontal position indicating the magnitude and direction of the feature's contribution to the predicted log-transformed N_2O flux. Point colors represent the underlying feature value for each observation, with red indicating higher values and blue indicating lower values. Consequently, a concentration of high-value points on the right-hand side of the plot suggests that larger values of the corresponding feature are associated with increased predicted emissions, whereas high-value points on the left-hand side indicate a negative association.

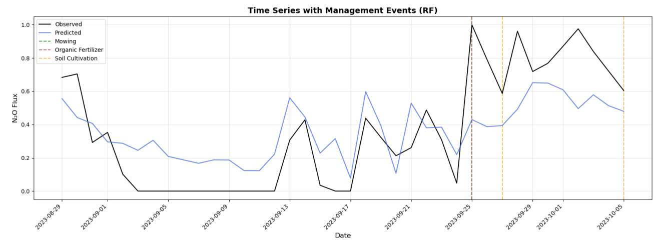
Across both datasets and model classes, several consistent patterns emerge. Net ecosystem exchange (NEE) is identified as one of the most influential predictors of N_2O emissions, appearing as the top-ranked variable in both XGBoost and Random Forest models for the Oensingen dataset, and among the top three predictors for both models trained on Chamau. In addition, nitrogen fertilization-related variables play a prominent role in the Chamau models. In contrast, fertilization variables do not appear as important predictors in the Oensingen models, which is expected given the absence of fertilization events in the corresponding test period.

In order to better understand common predictors of N_2O flux across different fields, we aggregate SHAP-based feature rankings across all trained models and feature settings for the Chamau and Oensingen 2021-23 datasets. Because the available predictors differ across feature settings, results are first grouped by model setting (base, lag, and augmented lag). Within each group, predictors are ranked according to how frequently they appear among the top 3, 5, and 10 features based on mean absolute SHAP values across models (XGBoost and Random Forest), datasets (Chamau and Oensingen 2021-23) and temporal resolutions (hourly and daily). In the tables below, we report the proportion of models in which a feature appears among the top-K ranked predictors and present only the top five predictors for clarity.

Looking at the tables, a few variables appear to consistently emerge as influential predictors across datasets, models, and feature settings. These predictors can be grouped into three categories: nitrogen fertilization variables (such as the half life decay of nitrogen fertilizer and days-since-fertilization indicators), net ecosystem exchange (NEE), and soil properties such as soil temperature and moisture at shallow depths.



(a) Chamau – Random Forest predictions with management events.



(b) Oensingen – Random Forest predictions with management events.

Figure 1: Observed and predicted N_2O flux time series for the Random Forest model, overlaid with key agricultural management events. Vertical dashed lines indicate the timing of mowing, fertilization, grazing, and soil cultivation events.

Nitrogen fertilization variables are particularly prominent in lagged and augmented-lag feature settings, where temporally decayed fertilizer inputs frequently appear among the top-ranked predictors. This pattern indicates that incorporating temporal structure allows the models to capture delayed effects of nitrogen applications on predicted N_2O emissions. NEE and its lagged variants, exhibit high robustness across all feature settings. Soil moisture and temperature variables also appear consistently among the top predictors.

Table 1: Robustness of SHAP feature importance (base)

Feature	Top-3 (%)	Top-5 (%)	Top-10 (%)
NEE	62.5	100.0	100.0
RECO	50.0	100.0	100.0
SoilWater_5cm	50.0	75.0	100.0
SoilTemp_15cm	50.0	50.0	100.0
SoilTemp_30cm	50.0	50.0	100.0

Table 2: Robustness of SHAP feature importance (lag)

Feature	Top-3 (%)	Top-5 (%)	Top-10 (%)
Fert_N_kg_ha_expHL7d	57.1	85.7	85.7
Fert_N_kg_ha_expHL14d	42.9	42.9	71.4
NEE	28.6	71.4	85.7
Fert_N_kg_ha_expHL3d	28.6	57.1	71.4
SoilWater_5cm	28.6	28.6	57.1

Table 3: Robustness of SHAP feature importance (augmented lag)

Feature	Top-3 (%)	Top-5 (%)	Top-10 (%)
NEE_roll7d_mean	66.7	83.3	83.3
Fert_N_kg_ha_expHL14d	50.0	50.0	66.7
DaysSince_FertOrganic	50.0	50.0	50.0
Fert_N_kg_ha_expHL7d	33.3	66.7	100.0
SoilWater_5cm	33.3	33.3	66.7

6 Discussion

6.1 Challenges

One of our biggest challenges was data leakage in some of our experiments: when running our SHAP values analysis pipeline, we first checked for data leakage to ensure our results were valid, and we detected some significant leakage in our XGBoost experiments, therefore invalidating our experiments. To address this, we implemented the leakage check described in the methodology section and reran the XGBoost experiments to obtain valid results.

Another challenge we encountered was related to some inconsistencies and errors in our initial data analysis and preprocessing. Although not major, these issues were pointed out to us by our challenge giver, and we took the necessary steps to address them, update our analysis plots and rerun our experiments using the corrected datasets.

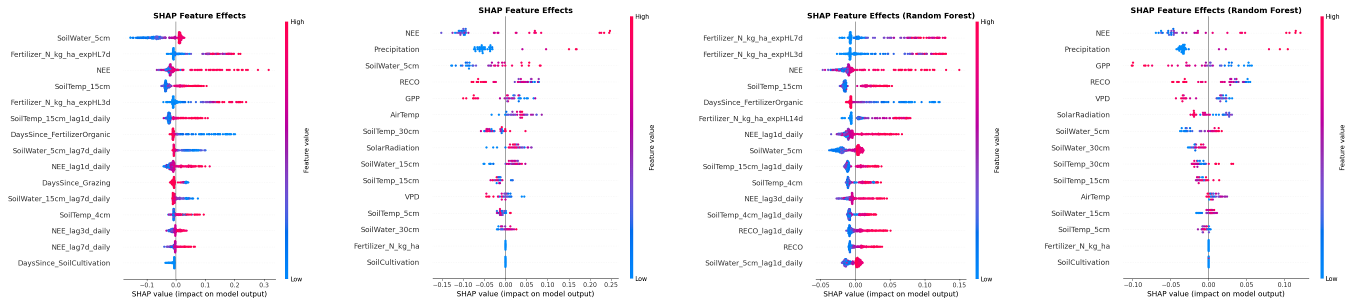
6.2 Limitations

*6.2.1 **Data Quality**.*

The predictive performance and interpretability of our models are hindered by several data quality limitations. Firstly, the majority of the available datasets cover relatively short temporal periods and often contain large gaps. This restricts the models' ability to learn long-term dynamics, such as seasonal patterns and delayed responses to environmental drivers. Moreover, small sample sizes increase the susceptibility of the models to overfitting, while reduced test set sizes may lead to seemingly favorable results occurring by chance. Overall, these constraints limit the robustness and generalizability of site-specific model outcomes.

Secondly, a considerable proportion of key predictor variables are gap-filled or model-derived rather than directly observed. For example, in the Chamau dataset, net ecosystem exchange (NEE), gross primary production (GPP), and ecosystem respiration (RECO) are imputed using Random Forest, which is particularly concerning given NEE was identified as one of the most influential predictors of N_2O emissions. Similarly, in the Forel dataset, soil water content and soil temperature variables at multiple depths were gap-filled using XGBoost. As these predictors originate from secondary models that may themselves introduce bias, their use adds additional uncertainty and may propagate assumptions from the imputation models into the downstream N_2O prediction task.

Finally, in the Chamau dataset, key variables such as NEE, GPP, and RECO are assigned to individual fields based on wind direction.



(a) Chamau – XGBoost (Daily Lag)

(b) Oensingen – XGBoost (Daily Base)

(c) Chamau – Random Forest (Daily Lag)

(d) Oensingen – Random Forest (Daily Base)

Figure 2: SHAP beeswarm plots for the best-performing XGBoost and Random Forest models trained on the Chamau and Oensingen datasets. Features are ordered by mean absolute SHAP value. Each point represents a test observation, with colors indicating the corresponding feature value.

While this provides a practical approach for field-level attribution, errors in wind direction can lead to the misallocation of these variables across fields, introducing additional noise and uncertainty into the dataset. Given that Chamau is the largest dataset and plays a central role in our study, the presence of these additional sources of uncertainty is particularly relevant for interpreting our results.

6.2.2 *SHAP-based explainability*.

One issue with utilizing SHAP values to determine key drivers of N_2O emissions is that they quantify variable importance as learned by the models and do not imply causal relationships between predictors and real-world emission processes. As a result, features identified as important by SHAP should be interpreted as influential within the model's predictive structure rather than as direct drivers of physical N_2O emissions.

A further limitation is that SHAP values are derived from the model's predictions and are therefore constrained by the predictive performance of the underlying model. When model performance is moderate, a substantial proportion of the variance in N_2O emissions remains unexplained. In this study, model coefficients of determination generally range between 0.3 and 0.6, and while SHAP values provide useful insight into which variables contribute most strongly to model predictions, they may not fully reflect the true importance of these variables in governing N_2O emissions in real-world systems.

6.2.3 *Project Timeline*.

One last limitation is that, although we managed to address all challenges outlined in subsection 6.1, they ultimately cost us a significant amount of time, as we had to rerun most of the notebooks to ensure everything included in our public repository, as well as our experimental results, were valid. Given the tight timeline of the project, the lost time could have been spent exploring some of the proposed directions for future work or addressing some of the other acknowledged limitations of our work.

7 Future Work

One direction for further research could involve running our hyperparameter optimisation pipeline on models that only use the most significant features extracted using our SHAP values analysis. Given that, in our experiments, adding more predictors to the model did not always result in an increase in predictive performance, this approach could strike a balance between having enough predictors to learn the more intricate patterns of N_2O fluxes whilst also ensuring against overfitting.

Another direction could be to combine the datasets (and therefore amass more data) and then test the existing models on these combined datasets. The main approach we were considering would be to introduce an additional 'field' variable indicating the original site the data came from, and then perform the train-test split using both the 'date' and 'field' variable (e.g., time based 80:20 split where we include the first 80% of the data in each field in the training set and the remaining 20% from later time periods in the test set).

8 Conclusion

In this study, we applied several machine learning models to predict N_2O emissions across multiple agricultural sites in Switzerland, combining explicit temporal feature engineering with SHAP-based interpretability. Tree-based ensemble methods, particularly Random Forest and XGBoost, consistently outperformed ElasticNet, SVM, and neural network models, with daily-aggregated data yielding the strongest performance. SHAP analysis highlighted net ecosystem exchange, soil moisture/temperature, and nitrogen fertilization related variables as influential predictors, while also revealing substantial site-specific differences driven by data availability and management records. Although overall predictive performance remained moderate, our results demonstrate that interpretable machine learning can capture meaningful temporal patterns in N_2O emissions and provide useful insights into dominant drivers, while underscoring the importance of data quality and the limits of purely data driven approaches.

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Appendices

A Dataset Structure

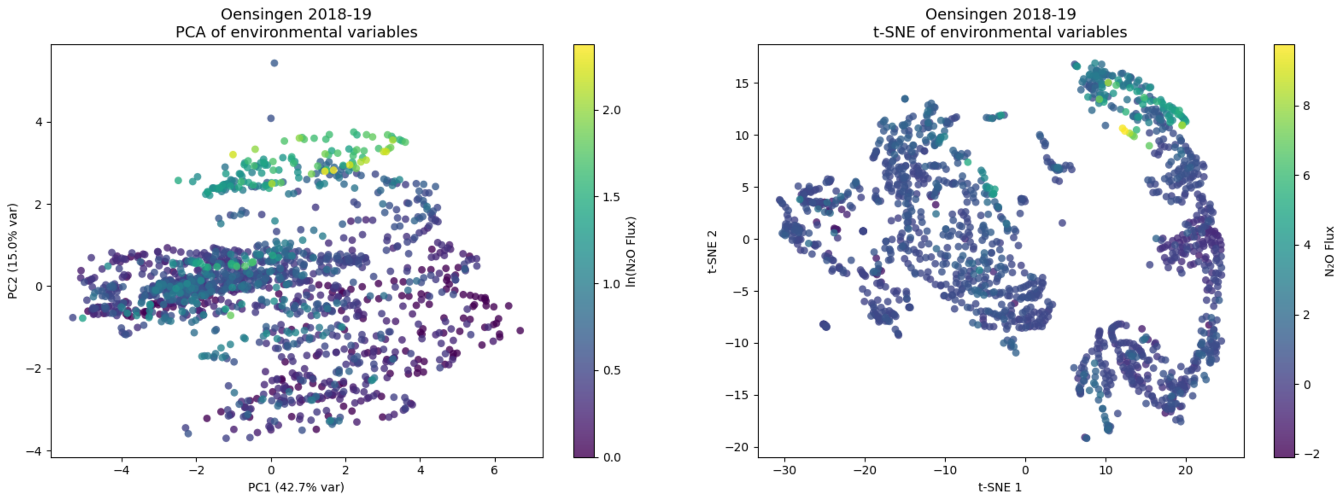

Figure 3: PCA plot (left) and t-SNE plot (right) showing more insights into the structure of the Oensingen 2018-19 dataset [12].

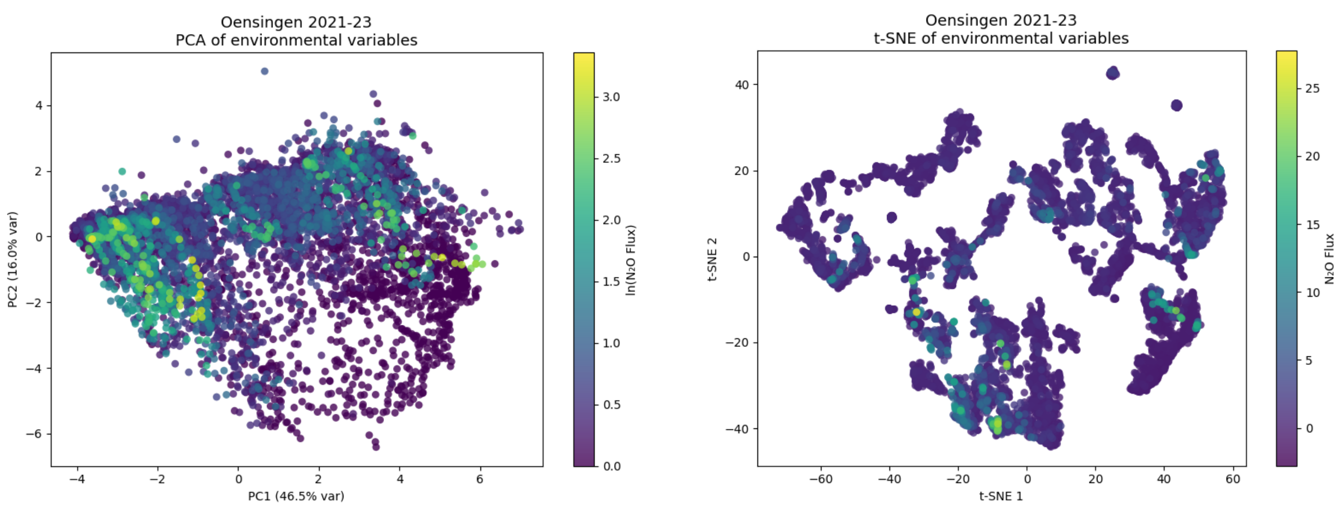

Figure 4: PCA plot (left) and t-SNE plot (right) showing more insights into the structure of the Oensingen 2021-23 dataset [14].

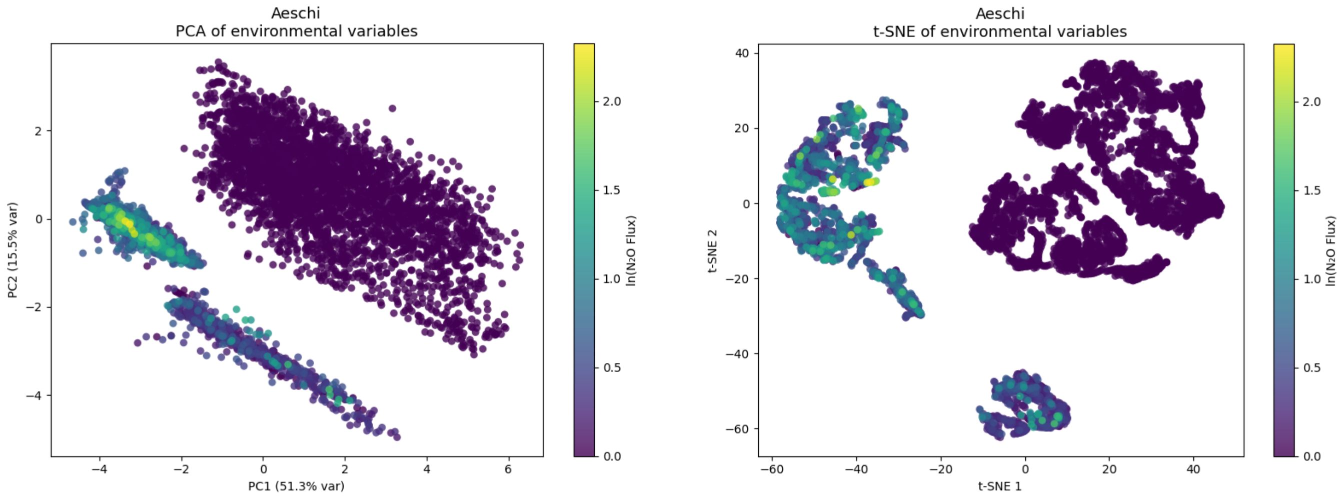

Figure 5: PCA plot (left) and t-SNE plot (right) showing more insights into the structure of the Aeschi 2018-20 dataset [12, 13].

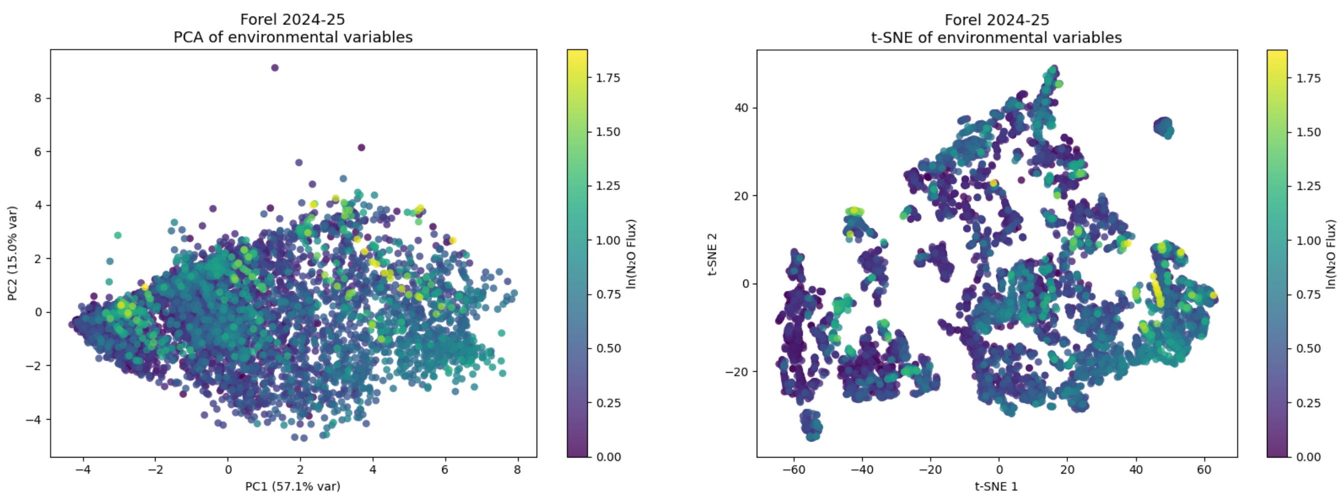

Figure 6: PCA plot (left) and t-SNE plot (right) showing more insights into the structure of the Forel 2024-25 dataset.

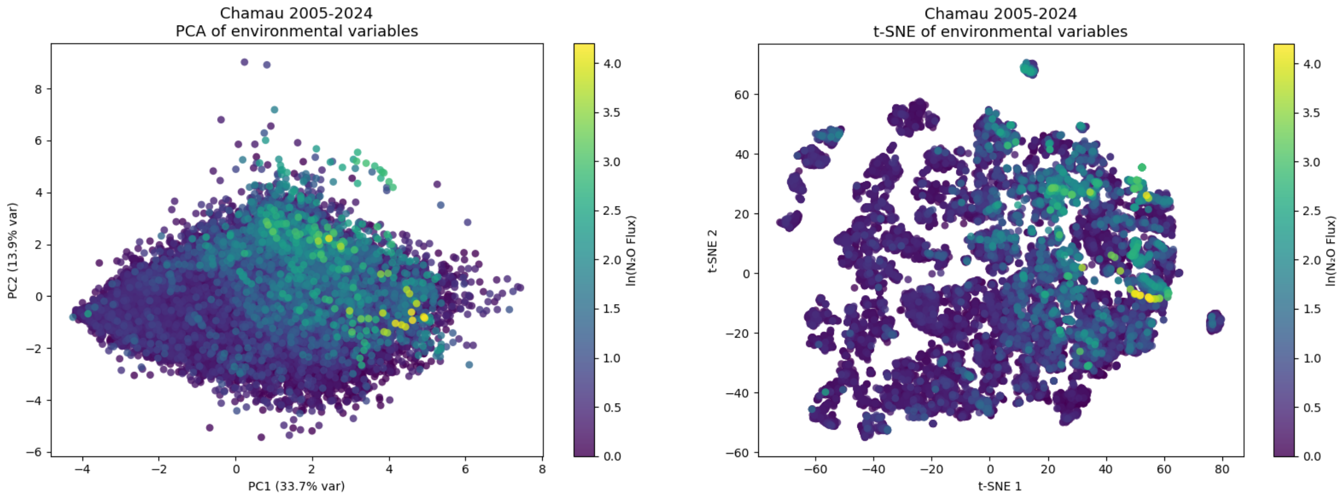

Figure 7: PCA plot (left) and t-SNE plot (right) showing more insights into the structure of the Chamau 2005-24 dataset [11].

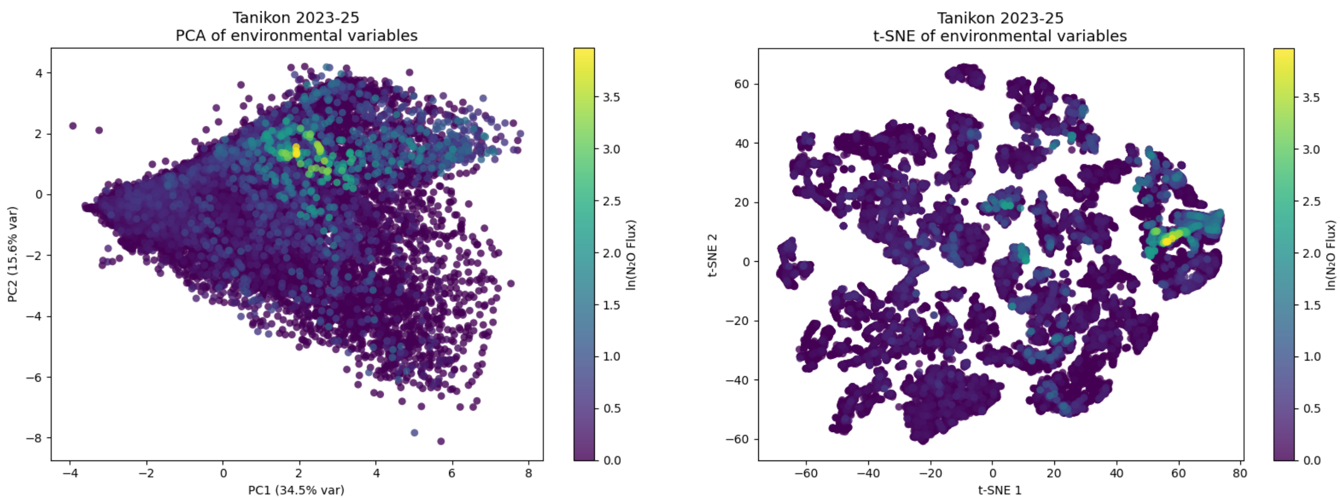

Figure 8: PCA plot (left) and t-SNE plot (right) showing more insights into the structure of the Tänikon 2023-25 dataset.

B Experiment Results

In this appendix we provide a more detailed breakdown of the results for each of the 5 models used in our experiments. We include the results for each of the 6 experimental settings used only for the top 3 best performing datasets.

B.1 ElasticNet

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.067
Hourly (lag predictors)	-0.092
Daily (base predictors)	0.117
Daily (lag predictors)	0.137

Table 4: Observed results for the Chamau dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.054
Hourly (lag predictors)	-0.434
Daily (base predictors)	-0.029
Daily (lag predictors)	0.031

Table 5: Observed results for the Oensingen 2021-23 dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	-0.343
Hourly (lag predictors)	-0.484
Daily (base predictors)	-1.062
Daily (lag predictors)	-1.467

Table 6: Observed results for the Oensingen 2018-19 dataset.

B.2 Random Forest

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.230
Hourly (lag predictors)	0.330
Hourly (augmented predictors)	0.325
Daily (base predictors)	0.171
Daily (lag predictors)	0.371
Daily (augmented predictors)	0.337

Table 7: Observed results for the Chamau dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.034
Hourly (lag predictors)	0.195
Hourly (augmented predictors)	0.140
Daily (base predictors)	0.531
Daily (lag predictors)	0.320
Daily (augmented predictors)	0.331

Table 8: Observed results for the Oensingen 2021-23 dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	-0.865
Hourly (lag predictors)	0.111
Hourly (augmented predictors)	0.220
Daily (base predictors)	-0.820
Daily (lag predictors)	-0.187
Daily (augmented predictors)	-0.108

Table 9: Observed results for the Forel dataset.

B.3 XGBoost

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.341
Hourly (lag predictors)	0.218
Hourly (augmented predictors)	0.092
Daily (base predictors)	0.059
Daily (lag predictors)	0.300
Daily (augmented predictors)	0.480

Table 10: Observed results for the Chamau dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.060
Hourly (lag predictors)	0.205
Hourly (augmented predictors)	0.046
Daily (base predictors)	0.525
Daily (lag predictors)	0.347
Daily (augmented predictors)	0.162

Table 11: Observed results for the Oensingen 2021-23 dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.070
Hourly (lag predictors)	0.281
Hourly (augmented predictors)	0.289
Daily (base predictors)	-0.016
Daily (lag predictors)	0.231
Daily (augmented predictors)	0.270

Table 12: Observed results for the Oensingen 2018-19 dataset.

B.4 Multilayer Perceptron

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.055
Hourly (lag predictors)	0.199
Hourly (augmented predictors)	0.227
Daily (base predictors)	0.005
Daily (lag predictors)	-0.266
Daily (augmented predictors)	-0.182

Table 13: Observed results for the Forel dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.143
Hourly (lag predictors)	-0.092
Hourly (augmented predictors)	-0.260
Daily (base predictors)	0.348
Daily (lag predictors)	-1.072
Daily (augmented predictors)	-0.073

Table 14: Observed results for the Oensingen 2021-23 dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.213
Hourly (lag predictors)	-0.004
Hourly (augmented predictors)	0.203
Daily (base predictors)	0.012
Daily (lag predictors)	-0.660
Daily (augmented predictors)	-0.017

Table 15: Observed results for the Oensingen 2018-19 dataset.

B.5 Support Vector Machine

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.120
Hourly (lag predictors)	0.334
Hourly (augmented predictors)	0.324
Daily (base predictors)	-0.086
Daily (lag predictors)	0.287
Daily (augmented predictors)	0.279

Table 16: Observed results for the Chamau dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	-0.401
Hourly (lag predictors)	-0.318
Hourly (augmented predictors)	-0.410
Daily (base predictors)	0.437
Daily (lag predictors)	0.515
Daily (augmented predictors)	0.423

Table 17: Observed results for the Oensingen 2021-23 dataset.

Experiment	Accuracy (R^2)
Hourly (base predictors)	0.111
Hourly (lag predictors)	-0.001
Hourly (augmented predictors)	0.156
Daily (base predictors)	0.177
Daily (lag predictors)	0.521
Daily (augmented predictors)	0.446

Table 18: Observed results for the Forel dataset.

C Hyperparameter Search Spaces

Model	Hyperparameter search space
Random Forest	`n_estimators`: [100, 1000] `max_depth`: [5, 50] `min_samples_split`: [2, 20] `min_samples_leaf`: [1, 15] `max_features`: {0.2, 0.25, 0.3, 0.35, 0.4, 0.5, sqrt, log2}
XGBoost	`learning_rate`: [0.01, 0.05, 0.1, 0.2] `max_depth`: {3, 4, 5, 6, 8} `n_estimators`: {100, 200, 300, 500} `subsample`: [0.6, 0.7, 0.8, 0.9] `colsample_bytree`: [0.6, 0.7, 0.8, 0.9] `min_child_weight`: {1, 3, 5} `gamma`: {0, 0.1, 0.2, 0.5}
Support Vector Machine	`kernel`: rbf `C`: [0.1, 100] (log scale) `epsilon`: [0.01, 0.3] `gamma`: {scale} ∪ [0.001, 0.1] (log scale)

Table 19: Hyperparameter search spaces used for each of the three models we ran Bayesian Optimisation on.